

Lab 1 – Chance Brock

1.)

a.) `x == y` True.

b.) `x is y` False.

c.) `z = x` True.

d.) `'=='` compares the values of the two objects. `'is'` compares the identity of said objects. `id(x) == id(z)` will always be true if neither are changed. Z was created to refer to the identity of x, so both are tied to the same object.

2.)

a.)

`a is b` True.

`c is d` False.

Python caches small integers as singleton objects. So, a and b refer to the same object. C and d are out of this range, so they do not fall into this caching.

b.) Using `'is'` to compare small integers is just a bug waiting to happen because it reacts differently to different integers. It can detect identity instead of numerical value.

3.)

a.) The “fail fast” philosophy favors immediately raising the `TypeError`. Silent coercion is dangerous for a Student ID lookup table because IDs have strict formatting rules.

4.)

a.) `{"CS311": "A", "CS215": "B", "CS258": "C"}`

b.) This happens because Python evaluates default argument expressions once, when the function is defined. So `gradebook` isn't being reset every time the function is called and instead is being added onto.